

09/06/26 "DAY 14" kamineni Dhana sri
Reethi

01 Working on

■ Data collection - google sheets

↳ Available in github repository
(TIIIT Nagpur outreach)
↳ drive.

02 Working on

■ Matlab. code: *MatlabCode-01 P.mpl*

01 3 Models

↳ Defect free Eq (1) to Eq (8)

↳ Defected (Cn) Eq (9) to (10)

↳ Defected (RLn) Eq (11) to (15)

02 Code.

→ Part 01

clc;
clear;

close all;

→ Part 02

%. constants

%. constants

→ part 03

%. constants

%. Input parameters

%. constants

→ part 04

%. constants ϵ_0, μ_0, q

%. Material parameters $\rho_0, \alpha, \mu_{in}, \epsilon_{pr}, \epsilon_{pr}, \epsilon_{pr}$

%. constants

→ part 05

%. constants

%. Equation (1) to (15)

→ part 06

%. constants

%. display results

%. constants

data excel connect.
pdf oip / github.
eq matlab.
day 13 updated written notes
also take prg
day 13 notes along with edited
day 13 cu.

R L C C_{leak} G_{leak} R_{sh} L_{sh}
values display.

Values (yet to edit (not accurate))

01 Defect free

01 $R_{TSV} = 0.138079 \Omega$

02 $L_{TSV} = 3.495917 \text{ pH}$

03 $G_{si} = 5.901387 \times 10^{-4} \text{ S}$

04 $C_{ox} = 1.355716 \times 10^{-14} \text{ F}$

05 $C_{dep} = 4.342187 \times 10^{-16} \text{ F}$

06 $C_{si} = 6.217855 \times 10^{-15} \text{ F}$

07 $G_{dep} = 3.027788 \times 10^{-4} \text{ S}$

08 $I_{sub} = -5.993566 \times 10^{-4} \text{ A}$

09 $C_{dep} = 7.167292 \times 10^{-15} \text{ F}$

10 $C_{si} = 6.217855 \times 10^{-15} \text{ F}$

02 Defected (Cn)

$G_{leak} = 2.386061 \times 10^{-14} \text{ F}$

$R_{leak} = 7.391138 \times 10^{-12} \text{ H}$

03 Defected (RL)

$R_{sh} = 0.00833 \Omega$

$L_{sh} = 7.391138 \times 10^{-12} \text{ H}$

Working on

■ Hspice L network analysis.

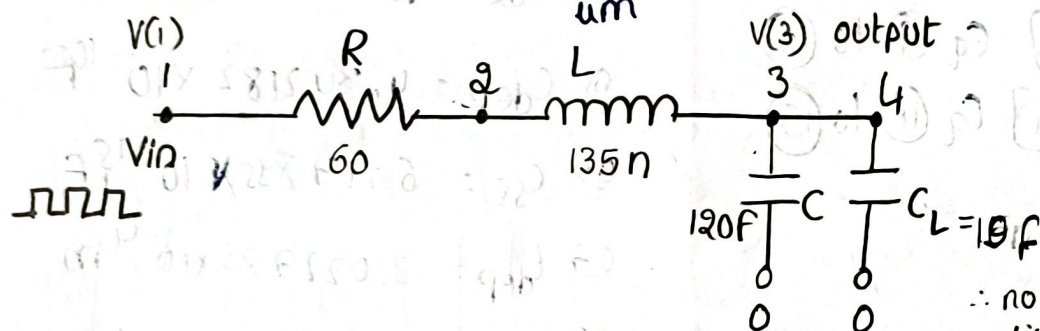
▲ Workspace 01 → Applications. [Top left corner] → Accessories → Text editor. [code writing]

▲ Notes of Interconnect length = 500 μm

$$\text{Resistance} = 0.12 \frac{\Omega}{\mu\text{m}} \Rightarrow 0.12 \times 500 = 60 \Omega$$

$$\text{Inductance} = 270 \frac{\text{pH}}{\mu\text{m}} \Rightarrow 270 \times 500 = 135 \text{ nH}$$

$$\text{Capacitance} = 0.24 \frac{\text{fF}}{\mu\text{m}} \Rightarrow 120 \text{ fF}$$



voltage → nodes
current → element.

∴ no need to mention units like Ω , H, F in this!

▲ Text editor — Code.

*** L Network ***

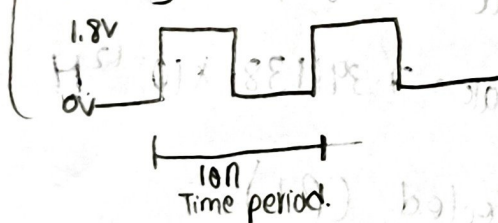
.tema5

*** Supply voltage ***

Vin 1 0 pulse (0 1.8 0s 100p 100p 5n 10n)

N	+ve GND node node	low level V	High level V	t_d	t_r	t_f	Pulse width	TP
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Explaining:



→ at $t=0$ $V=0V$

$t=100\text{ps}$ $V=1.8V$ Stays as 1.8V for 5nS

$t=200\text{ps}$ $V=0V$ stays as 0V for 5nS.

this is complete cycle

$$F = \frac{1}{TP} = \frac{1}{10n} = 100 \text{ MHz}$$

t_d = delay time

t_r = rise time

t_f = fall time

TP = Total time Period.

N = Name of voltage source.

$$\frac{PW}{TP} = \frac{5nS}{10nS} = 50\% \text{ duty cycle}$$

*** Network ***

R	1	2	60
L	2	3	5n
C	3	0	120F
CL	3	0	10F

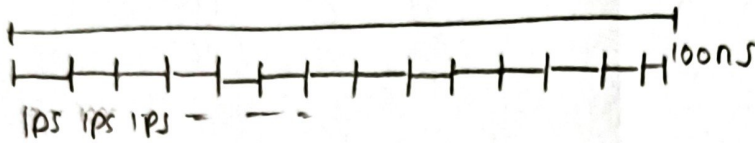
all Resistances R_1, R_2 if more than one present
RLGC --

Component $\begin{matrix} 1 & 2 \\ \text{btw which nodes} & \end{matrix}$ $\begin{matrix} 60 \\ \text{its value} \end{matrix}$
it is present

*** operation part ***

- option post (Run simulation + save waveforms)
- trans 1p 100n. (transient analysis T_{step} T_{stop})

1ps $\rightarrow$ saves result every 1 picosecond
 100n $\rightarrow$ Run simulation unit 100 nanosec



*** Measure ***

- Measure tran tr trig v(1) val=0.9 rise=1 targ v(3) val=0.9 rise=1
- Measure tran tf trig v(1) val=0.9 fall=1 targ v(3) val=0.9 fall=1
- Measure tran Pavg AVG par('v(1)*I(vin)') from 0 to 100n.

*** plot

- print v(3) I(L) I(c)
- end.

measure tran. tr

(create a measurement name tr. and measure it during a transient simulation).

trig $\rightarrow$ (trigger $\Rightarrow$ Input)

trig v(1) val=0.9 rise=1 (start timing when node crosses 0.9V on the first rising edge).

targ v(3) val=0.9 fall=1 (stop timing when node 3 crosses 0.9V on the first falling edge.)

$$t_r = t_{v(3)=0.9V} - t_{v(1)=0.9V}.$$

print v(3) $\rightarrow$ measure $V_{u.r.t T}$ $\rightarrow$ voltage at node 3 & I flowing in inductor.

Example	Time	V(3)	I(L)
	0n	0V	0mA
	1n	0.5V	2mA
	2n	1.8V.	uMA.

- end of code.

$\therefore$ waveforms in the oip-keerthi folder.

(after this process continuation is in day 15.